MZUZU UNIVERSITY

ICT

Diploma in ICT

MATH 1101-Precalculus

End of Semester Exam

Friday, 15 September, 2017

Time Allowed: 3 hours (14:00 - 15:00)

INSTRUCTIONS:

- Check that the paper contains **11 questions**.
- Attempt **all** questions.
- Show all your working clearly.
- Use proper notation
- Simplify your answers as far as possible
- **NO CALCULATORS ALLOWED**

1. Find the perimeter of the triangle formed by joining the mid-points of the sides of ΔABC with vertices $A(1,1)$, $B(9,3)$ and $C(3,5)$.

[5 marks]

2. (a) Find all real solutions for $\sqrt{y+2} = y - 4$.

   (b) Solve the equation $x^2 + (\sqrt{2}-1)x = \sqrt{2}$, leaving your answer in surd form.

[5/6 marks]

3. (a) Simplify

$$\frac{x + \dfrac{xy}{y-x}}{\dfrac{y^2}{x^2-y^2} + 1}.$$

   (b) Show that

$$\frac{\left(a+\frac{1}{b}\right)^a \left(a-\frac{1}{b}\right)^b}{\left(b+\frac{1}{a}\right)^a \left(b-\frac{1}{a}\right)^b} = \left(\frac{a}{b}\right)^{a+b}.$$

[6/7 marks]

4. Let $f(x) = \sqrt{\dfrac{4x-3}{x^2-4}}$. Find the domain of $f(x)$.

[8 marks]

5. Solve the following:

   (a) $3^{5x-8} = 9^{x+2}$

   (b) $5^{2x-1} = 6^{x-2}$

[4/5 marks]

6. (a) Factor $\tan^2 A + 5 \cdot \tan A + 6$

   (b) Complete the following table:

θ	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{3\pi}{2}$	2π
$\sin\theta$							
$\cos\theta$							

[3/7 marks]

7. A 100m vertical antenna is on the roof of a building. From a point on the ground, the angles of elevation to the top and bottom of the antenna are 60° and 45° respectively. Find the height of the building.

[10 marks]

8. Solve the following:

   (a) $x^2 - 16 \geq 0$

   (b) $x^4 - 6x^2 + 4 = 0$

[3/5 marks]

9. Suppose that in 1995, the population of a certain country was 2 million and increasing with a relative growth rate of 2% per year. By sketching the graph, estimate the year in which the population will reach 3 million. Verify your answer algebraically.

[10 marks]

10. (a) State the amplitude, period, phase shift and vertical shift of the following function:

$$f(x) = 3 \sin\left(2x - \frac{2\pi}{3}\right)$$

   (b) Hence, sketch the graph of

$$f(x) = 3 \sin\left(2x - \frac{2\pi}{3}\right)$$

[4/6 marks]

11. (a) Convert $\frac{11\pi}{6}$ radians to degrees.

   (b) Convert 450° to radians.

[3/3 marks]

End of the question paper